Day-42

SSH

Consde Port Security - login

- By default, no password is needed to access the CLI of a Cisco IOS device via the console port.

- We can configure a password on the console line.
  ↳ A user'll have to enter to a password to access the CLI via the console port.

i) R1(config) # line console 0 [There's only a single console line, so the number is always 0]
- single console line means there can only be a single console connection at once. No more than one people can configure device at same time via separate console ports.

ii) R1(config-line) # password ccnc [configure console line's password]
iii) R1 (config-line) # login [tells the device to require a user to enter the configured password to access the CLI via console port]

iv) R1 (config-line) # end
v) R1 # exit
} this just not needed. kei faral pardaine. We configured & logout bhneya dekhaiya

The password ain't displayed when we type it.

Console Port Security - login local

Alternatively, we can configure the console line to require users to login using one of the configured usernames on the device.

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i) R1(config) # username jeremy secret ccnp

ii) R1(config) # line console 0

iii) R1(config-line) # login local → [tells the device to require a user to login using one of the configured usernames on the device]

iv) R1(config-line) # login local

So, login = pwd only & login local = username & pwd since we changed from login to login local, so console line's pwd for ccna can no longer be used. Now username & pwd only

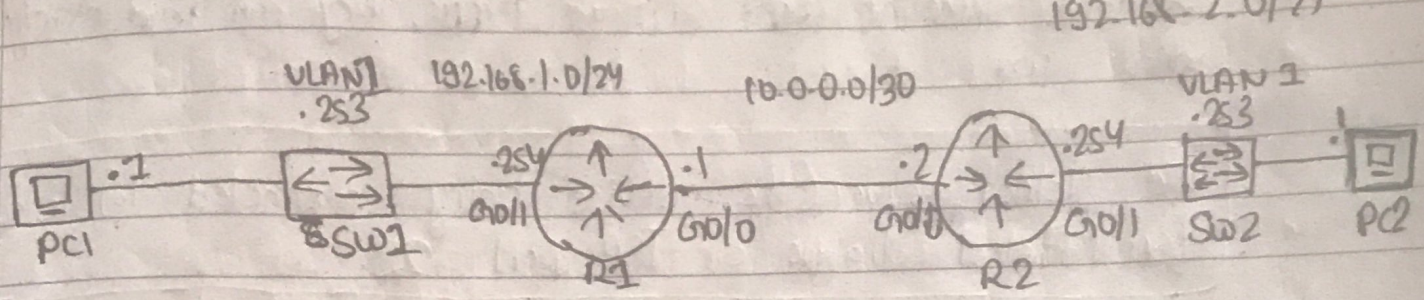
R1(ll) # exec-timeout 3 30
Log the user out after 3 mins. & 30 secs of inactivity

Layer 2 Switch - Management IP

Routers & layer 3 switches have IP addresses we can use to connect remotely to manage the devices. What about layer 2 switch?

- Layer 2 switches don't perform packet routing & don't build a routing table. They aren't IP routing aware.
- However, we can assign an IP address to an Switch Virtual Interface (SVI) to allow remoted connection to the CLI of the switch (using telnet or SSH).

Network Topology



The network admin is using PC2 & s/he needs to be able to connect to all of the devices in network to configure them. To allow this, layer 2 switches need IP address.

Configuration of layer 2 switch.

i) SW1(config) # interface vlan1
ii) SW1 (config-if) # ip address 192.168.1.253 255.255.255.0
iii) SW1 (config-if) # no shutdown
iv) SW1 (config-if) # exit

↳ Configure the IP address on the SVI in the same way as on a multilayer switch. Enable the interface if necessary

v) SW1(config) # ip default-gateway 192.168.1.254.
Configure the switch's default gateway. In this case, PC2 isn't in the same LAN as SW1. If SW1 doesn't have default gateway, it can't communicate with PC2.

Telnet

- Telnet (Teletype Network) is a protocol used to remotely access the CLI of a remote host.
- Telnet was developed in 1969.
- Telnet has been largely replaced by SSH, which is more secure.

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- Telnet sends data in plain text. No Encryption! So, if someone uses captures traffic, they can read exactly what we sent

> The telnet server (the device being connected to) listens for Telnet traffic on TCP port 23.

Telnet client IS device that's connecting & uses random port number.

Telnet Configuration

- SW1(config)# enable secret ccna
  If an enable password/secret ain't configured, we won't be able to access privileged exec mode when connecting via telnet.
- SW1(config)# username jeremy secret ccna.
  - Pani login local garna ho. Not compulsion for telnet
- SW1(config)# access-list 1 permit host 192.168.2.1
  - to create restriction on vty lines so that only admin can access telnet. Not compulsion but better for security.
- SW1(config)# line vty 0 15
  Telnet/SSH is configured on the VTY lines. There are 16 lines available, so upto 16 users can be connected at once. (VTY stands for Virtual Tele Type)
  So line vty 0 15 means kind of int g0/1 or g0/1-15.
- SW1(config-line)# login local.
- SW1(...)# exec-timeout 5 0

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* Old version of Cisco IOS only had 5 VTY lines

VTY - Virtual Tele Type

- SW1 (config - line) # transport input telnet
  - only allows telnet. If telnet ssh allows both. If input none allows none.

- SW1 (config-line) # access-class 1 in
  - * access-class - applies ACL to VTY lines
  - * ip access-group - applies ACL to an interface

Secure Shell (SSH).

- SSH (Secure Shell) was developed in 1995 to replace less secure protocols like Telnet.

  In computing, a shell is a computer program which exposes an operating system's services to human user or other program.

- SSHv2, a major revision of SSHv1, was released in 2006
- If a device supports both SSHv1 & SSHv2, it's said to run version 1.99.

- Provides security features such as data encryption & authentication
- The SSH server listens for SSH traffic on TCP port 22

SSH Configuration: Check SSH support

Before configuring for SSH, make sure that the version of IOS on your ~~dev~~ device supports it.

SW1 # show version

..... (vios-17-ADVENTERPRISE**K9**-M)

IOS images that support SSH will have k9 in their name.

- Cisco Exports NPE (No Payload Encryption) IOS images to countries that've restriction on encryption technology.
- NPE IOS images don't support cryptographic features such as SSH.

- SW1# show ip ssh
  SSH Disabled - version 1.99.
  [If ssh isn't supported its mentioned here

SSH Configuration: RSA Keys

- To enable and use SSH, we must generate an RSA public & private key pair.
- The keys are used for data encryption/decryption, authentication, etc.
- SW1(config)# ip domain name jeremysitlab.com
  FQDN = Fully Qualified Domain Name (host name + domain name)
  The FQDN of device is used to name the RSA keys
- SW1(config)# crypto key generate rsa
  How many bits in modulus [512]: 2048
  Length must be greater than 768 for SSHv2.
  Alternate is
  crypto key generate rsa modulus length
- SW1(config)# do show ip ssh
  SSH Enabled - version 1.99

SSH Configuration: VTY Lines

- Sw1(config)# enable secret ccna.
- Sw1(config)# username jeremy secret ccna.
- Sw1(config)# access-list 1 permit host 192.168.2.1
- Sw1(config)# ip ssh version 2 [optional but recommended, restricts SSH to version 2 only]
  &
- Sw1(config)# line vty 0 15.
- Sw1(config-line)# login local. [We can't use just login. to username is is must. So login local (optional but recommended)]
- Sw1(config-line)# exec-timeout 5 0
- Sw1(config-line)# transport input ssh [allows ssh only no telnet or other]
  Sw( )# access-class 1 in

SSH configuration.

1) Configure host name
2) Configure DNS domain name
3) Generate RSA key pair!
4) Configure enable PW, username/PW
5) Enable SSHv2 (only).
6) Configure VTY lines

(Configuring ssh is asked in CCNA exam)

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